

Generation Interconnection System Impact Study Report

For Queue Project Q21-1720

Bpa Troutdale 100 kV

June 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1720. The Interconnection Customer proposes a 68.9 MW solar photovoltaic facility interconnecting at the Bpa Troutdale 100 kV substation in the Illinois area. The request was received on April 14, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 567 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$901,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bpa Troutdale – Sarival 100.0 kV	Lisle – Bpa Troutdale 100.0 kV L	115.2	241	100.0
Lisle – Bpa Troutdale 100.0 kV L	Sarival – Bpa Troutdale 100.0 kV	115.2	241	100.0
Tropic Isle – Lisle 100.0 kV Line	Sarval – Bpa Troutdale 100.0 kV	105.6	140	61.2
Brenas 9201 – Sarival 100.0 kV L	Lisle – Bpa Troutdale 100.0 kV	101.3	146	35.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bpa Troutdale – Sarival 100.0 kV	Rebuild line with higher-capacity conductor	\$1,431,000	23% (\$326,000)
Lisle – Bpa Troutdale 100.0 kV	Rebuild line with higher-capacity conductor	\$1,432,000	23% (\$326,000)
Tropic Isle – Lisle 100.0 kV Line	Reconductor line	\$459,000	23% (\$104,000)
Sarival – Brenas 9201 100.0 kV	Reconductor line	\$635,000	23% (\$145,000)
	Total allocated to project		\$901,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.